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SYSTEM OF CABLE AND CONNECTORS WITH INTEGRATED SENSORS

Abstract

An electrical connector assembly includes a first connector member, a second connector member configured for connection with the first connector member, a first insulator disposed at least partially in the first connector member and/or the second connector member, a second insulator disposed at least partially in the first connector member and/or the second connector member, and a circuit board configured to be disposed at least partially between the first insulator and the second insulator in a connected configuration of the electrical connector assembly. The circuit board may include an electrical characteristic sensor and/or a temperature sensor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of a patent application Ser. No. 17/922,770, filed Nov. 1, 2022, which is a National Stage Application of PCT/EP2021/025163, filed Apr. 29, 2021, which claims the benefit of U.S. Provisional Application Ser. No. 63/018,885, filed May 1, 2020, the disclosures of which are incorporated herein by reference in their entireties. To the extent appropriate, a claim of priority is made to each of the above disclosed applications.

TECHNICAL FIELD

[0002] The present disclosure generally relates to electrical connectors, including electrical connectors that may, for example, be used in connection with aircraft.

BACKGROUND

[0003] This background description is set forth below for the purpose of providing context only. Therefore, any aspect of this background description, to the extent that it does not otherwise qualify as prior art, is neither expressly nor impliedly admitted as prior art against the instant disclosure.

[0004] Some electrical systems include cables and wires connected via electrical connectors that may not provide sufficient functionality, may be complex, may not be configured for high voltages, may not monitor electrical wiring and interconnection systems (EWIS), and/or may not be efficient. For example, some aircraft electrical systems may include numerous, separate components that may be relatively heavy, large, and/or inefficient.

[0005] There is a desire for solutions/options that minimize or eliminate one or more challenges or shortcomings of electrical connectors and systems. The foregoing discussion is intended only to illustrate examples of the present field and is not a disavowal of scope.

SUMMARY

[0006] In embodiments, an electrical connector assembly may include a first connector member, a second connector member configured for connection with the first connector member, a first insulator disposed at least partially in the first connector member and/or the second connector member, a second insulator disposed at least partially in the first connector member and/or the second connector member, and/or a circuit board configured to be disposed at least partially between the first insulator and the second insulator in a connected configuration of the electrical connector assembly. The circuit board may include an electrical characteristic sensor and/or a temperature sensor.

[0007] With embodiments, An electrical system may include a cable, a first electrical connector assembly connected to a first end of the cable, the first electrical connector assembly including a first electrical characteristic sensor and a first temperature sensor, a second electrical connector assembly connected to a second end of the cable, the second electrical connector assembly including a second electrical characteristic sensor and a second temperature sensor, and an electronic control unit (ECU) connected to the first electrical characteristic sensor, the first temperature sensor, the second electrical characteristic sensor, and the second temperature sensor. The ECU may be configured to determine a leakage current according to a difference between a first current sensed via the first electrical characteristic sensor and a second current sensed via the second electrical characteristic sensor.

[0008] In embodiments, a method of operating an electrical system may include sensing an electrical characteristic and/or a temperature associated with a cable via an electrical characteristic sensor and/or a temperature sensor connected to a circuit board of the electrical connector assembly, comparing, via an electronic control unit (ECU), the sensed electrical characteristic

and/or temperature with a specified or predetermined range, and/or detecting an error if the sensed electrical characteristic and/or temperature is outside the specified or predetermined range.
[0009] The foregoing and other potential aspects, features, details, utilities, and/or advantages of examples/embodiments of the present disclosure will be apparent from reading the following description, and from reviewing the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] While the claims are not limited to a specific illustration, an appreciation of various aspects may be gained through a discussion of various examples. The drawings are not necessarily to scale, and certain features may be exaggerated or hidden to better illustrate and explain an innovative aspect of an example. Further, the exemplary illustrations described herein are not exhaustive or otherwise limiting, and are not restricted to the precise form and configuration shown in the drawings or disclosed in the following detailed description. Exemplary illustrations are described in detail by referring to the drawings as follows:

[0011] FIG. 1 is an exploded perspective view generally illustrating an embodiment of an electrical connector assembly in accordance with teachings of the present disclosure.

[0012] FIG. 2 is a partial cross-sectional view generally illustrating an embodiment of an electrical connector assembly in accordance with teachings of the present disclosure.

[0013] FIG. 3 is a partial cross-sectional view generally illustrating an embodiment of an electrical connector assembly in accordance with teachings of the present disclosure.

[0014] FIG. 4 is a top view generally illustrating an embodiment of a circuit board in accordance with teachings of the present disclosure.

[0015] FIG. 5 is a perspective view generally illustrating an embodiment of a circuit board in accordance with teachings of the present disclosure.

[0016] FIG. 6 is a perspective view generally illustrating an embodiment of a circuit board in accordance with teachings of the present disclosure.

[0017] FIG. 7 is an exploded perspective view generally illustrating an embodiment of a second connector member in accordance with teachings of the present disclosure.

[0018] FIG. 8 is a partial cross-sectional view generally illustrating an embodiment of an electrical system in accordance with teachings of the present disclosure.

[0019] FIG. 9 is a flow chart generally illustrating an embodiment of a method of operating an electrical system in accordance with teachings of the present disclosure.

DETAILED DESCRIPTION

[0020] Reference will now be made in detail to embodiments of the present disclosure, examples of which are described herein and illustrated in the accompanying drawings. While the present disclosure will be described in conjunction with embodiments and/or examples, they do not limit the present disclosure to these embodiments and/or examples. On the contrary, the present disclosure covers alternatives, modifications, and equivalents.

[0021] In embodiments, such as generally illustrated in FIG. 1, an electrical connector assembly **100** may include a first connector member **102** and/or a second connector member **104** that may be configured for connection with each other. The first connector member **102** may include a first housing member **106**, a first insulator **108**, and/or one or more first terminals **110**. The second connector member **104** may include a second housing member **112**, a second insulator **114**, and/or one or more second terminals **116**. The first terminals **110** and the second terminals **116** may be configured for connection with each other, such as upon connection of the first connector member **102** with the second connector member **104**.

[0022] With embodiments, such as generally illustrated in FIGS. 2 and 3, an electrical connector

assembly **100** may be configured to connect (e.g., electrically) a first cable **202** with a second cable **206**. The first cable **202** may include one or more first conductors **204** (e.g., a plurality of first conductors **204**). The second cable **206** may include one or more second conductors **208** (e.g., a plurality of second conductors **208**). The one or more first terminals **110** may be configured for connection with the one or more first conductors **204**. The one or more second terminals **116** may be configured for connection with the one or more second conductors **208**. For example and without limitation, connection of the first connector member **102** with the second connector member **104** may connect the first cable **202** with the second cable **206** via the first terminals **110** and the second terminals **116**.

[0023] In embodiments, such as generally illustrated in FIGS. **1**, **2**, and **3**, an electrical connector assembly **100** may include a circuit board **118**. The circuit board **118** may be disposed at least partially in the first connector member **102** and/or the second connector member **104**. Additionally or alternatively, the circuit board **118** may be disposed at least partially between (e.g., axially) the first insulator **108** and the second insulator **114**. In some circumstances, the first insulator **108** may include a first portion **120** and/or a second portion **122** that may be spaced and/or separate from the first portion **120**. The second portion **122** may be disposed at least partially between (e.g., axially) the first portion **120** and the second insulator **114**, such as in a connected configuration of the electrical connector assembly **100** (see, e.g., FIG. **2**). The circuit board **118** may be disposed at least partially between the first portion **120** and the second portion **122**. For example and without limitation, the circuit board **118** and the second portion **122** may be disposed at least partially between the first portion **120** and the second insulator **114**. With embodiments, such as generally illustrated in FIG. **3**, the circuit board **118** may be disposed such that the circuit board **118** is disposed radially within portions of the first connector member **102** and the second connector member **104**.

[0024] With embodiments, a circuit board **118** may include one or more of a variety of shapes, sizes, configurations, and/or materials. As generally illustrated in FIGS. **4**, **5**, and **6**, a circuit board **118** may include a generally circular and/or cylindrical configuration, such as if the first connector member **102** and/or the second connector member **104** include cylindrical configurations. A circuit board **118** may include one or more apertures **300** that may extend into and/or through (e.g., in an axial direction) the circuit board **118**. The one or more apertures **300** may be configured to at least partially receive and/or electrically connect with the one or more first terminals **110** and/or the one or more second terminals **116**. For example and without limitation, the apertures **300** may be configured as plated through holes (PTHs) that may be connected to respective traces **302** that may be disposed on the circuit board **118**.

[0025] While embodiments of an electrical connector assembly **100** are disclosed with a circuit board **118** disposed in an axial configuration, an electrical connector assembly **100** may, additionally or alternatively, include a circuit board **118** with a radial configuration. A circuit board **118** may be disposed in a variety of locations within an electrical connector assembly **100**.

[0026] In embodiments, such as generally illustrated in FIG. **4**, an electrical connector assembly **100** may include an electrical characteristic sensor **304** and/or a temperature sensor **306** that may be connected to a circuit board **118**. For example and without limitation, an electrical characteristic sensor **304** and/or a temperature sensor **306** may be disposed on and/or connected to a surface (e.g., an axial surface) of the circuit board **118**. Additionally or alternatively, an electrical characteristic sensor **304** and/or a temperature sensor **306** may be disposed in the electrical connector assembly **100** separately from the circuit board **118** and electrically connected thereto (e.g., via one or more conductors). The electrical characteristic sensor **304** may include one or more of a variety of shapes, sizes, and/or configurations. For example and without limitation, the electrical characteristic sensor **304** may include a current sensor (e.g., a Hall sensor, a shunt, etc.), a voltage sensor, a resistance sensor, and/or an impedance sensor, among others. The electrical characteristic sensor **304** may be configured to sense (e.g., detect, measure, monitor, obtain information

regarding, etc.) a current of a cable (e.g., the first cable **202** and/or the second cable **206**), and/or may be configured to sense one or more currents of individual conductors (e.g., the first conductors **204** and/or the second conductors **208**).

[0027] With embodiments, such as generally illustrated in FIG. 6, a circuit board **118** may include one or more pins **308** that may extend from a surface of the circuit board **118**, such as in a generally axial direction. The one or more pins **308** may, for example, be configured for connection with the first terminals **110** and/or the second terminals **116**.

[0028] In embodiments, such as generally illustrated in FIG. 7, a second connector member **104** may include a circuit board **118** (e.g., in addition to and/or instead of a circuit board **118** of the first connector member **102**). The second insulator **114** may include a first portion **126** and a second portion **128** that may be separate and/or spaced from each other, such as in an axial direction. The circuit board **118** may be disposed at least partially between (e.g., axially) the first portion **126** and the second portion **128**. The second portion **128** may be disposed at least partially between the first portion **126** and the first insulator **108**, such as in a connected configuration of the electrical connector assembly **100**. For example and without limitation, the circuit board **118** and the second portion **128** may be disposed at least partially axially between the first insulator **108** and the first portion **126** of the second insulator **114**.

[0029] In embodiments, a circuit board **118** may, for example and without limitation, be connected to a first insulator **108** and/or a second insulator **114** (or portions **120**, **122**, **126**, **128** thereof) via an adhesive **214** (see, e.g., FIG. 2).

[0030] With embodiments, a second connector member **104** (and/or a first connector member **102**) may include a grommet **130**, a seal **132**, and/or one or more retainers **134**. The grommet **130** and the seal **132** may be disposed at opposite ends of the second connector member **104**, which may include a second insulator **114** and/or a circuit board **118** being disposed axially between the grommet **130** and the seal **132**. The grommet **130** may be configured for connection with the second cable **206** and/or the one or more second conductors **208**, such as to retain and/or seal the second connector member **104** with the second cable **206**. The seal **132** may be configured to engage a corresponding face or seal of another connector member (e.g., the first connector member **102**). The grommet **130** and/or the seal **132** may, for example and without limitation, comprise silicone. The one or more retainers **134** may be configured for connection with and/or to retain the second terminals **116**, which may be connected (e.g., soldered, crimped, etc.) with the second conductors **208**.

[0031] In embodiments, such as generally illustrated in FIG. 8, an electrical system **400** may include a first electrical connector assembly **100**, a second electrical connector assembly **100'**, an electronic control unit (ECU) **402**, a first cable **202**, a second cable **206**, and/or a third cable **210**.

[0032] With embodiments, the second electrical connector assembly **100'** may include the same or a similar configuration as the first electrical connector assembly **100**. For example and without limitation, the second electrical connector assembly **100'** may include a first connector member **102'**, a second connector member **104'**, a first insulator **108'**, a second insulator **114'**, and/or a circuit board **118'**.

[0033] In embodiments, an ECU **402** may be connected (e.g., via wired and/or wireless connection) with the circuit board **118** and/or the circuit board **118'**. For example and without limitation, the ECU **402** may be connected with an electrical characteristic sensor **304** and/or a temperature sensor **306** of the circuit board **118** via one or more of the first conductors **204**, one or more first terminals **110**, and/or one or more traces **302** of the circuit board **118**. Additionally or alternatively, the ECU **402** may be connected with an electrical characteristic sensor **304** and/or a temperature sensor **306** of the circuit board **118'** via one or more third conductors **212** of the third cable **210**.

[0034] In embodiments, an electrical system **400** may be configured to detect and/or respond to one or more of a variety of errors. Responding to an error may, for example, include shutting down and/or disconnecting a component connected to the cable **202**, **206** or conductor **204**, **208** with the

error. Errors may include, for example and without limitation, over/under voltage, over/under current, series/parallel arc faults, partial discharges, ground faults, and/or other faults. Errors in an electrical system **400** may be caused by one or more of a variety of issues, such as, for example, wire insulation vibration, chafing, aging, degradation, heat damage, delamination, cracking, and/or moisture penetration, among others. Errors may include a conductor **204**, **208** being damaged or breaking, and the electrical system **400** may be configured to detect a corresponding increase in resistance of the cable **202**, **206**. Additionally or alternatively, errors may, in some circumstances, result in some electrical current being leaked to a ground (e.g., a vehicle/aircraft chassis ground). If the leakage current is sufficiently high, an arc or short may be created, which may damage one or more components of the electrical system **400** or systems connected or proximate the electrical system **400**.

[0035] With embodiments, the ECU **402** may be configured to detect leakage current. For example and without limitation, the ECU **402** may obtain a first current via an electrical characteristic sensor **304** of a first electrical connector assembly **100** connected to a first end of a first cable **202**, and obtain a second current via an electrical characteristic sensor **304** of a second electrical connector assembly **100'** connected to a second end of the first cable **202**. If the difference between the first current and the second current is not within a leakage current threshold, the ECU **402** may indicate an error has occurred, which may allow for the cable **202**, **206** to be repaired or replaced before a potentially larger issue (e.g., arc, short, etc.) occurs. The leakage current threshold may depend, at least to some extent, on a temperature proximate the first cable **202** (e.g., a resistance of the first cable **202** may vary with temperature). The ECU **402** may select the leakage current threshold according, at least in part, to a first temperature sensed via a first temperature sensor **306** of the first electrical connector assembly **100** and/or a second temperature sensed via a second temperature sensor **306** of the second electrical connector assembly **100'**.

[0036] In embodiments, a circuit board **118** and/or components connected thereto (e.g., an electrical characteristic sensor **304**, a temperature sensor **306**, etc.) may be electrically connected with the first terminals **110** and/or the second terminals **116**. For example, the circuit board **118**, the electrical characteristic sensor **304**, and/or the temperature sensor **306** may be configured to draw power from and/or provide information/data to (e.g., via power line communication or "PLC") the conductors **204**, **208** via the terminals **110**, **116**. The provided information/data may be transmitted via the conductors **204**, **208** to the ECU **402**, for example. The circuit board **118** may, for example, include one or more current loops **312** to facilitate connections between the circuit board **118**, the electrical characteristic sensor **304**, and/or the temperature sensor **306** for obtaining power and/or providing information (e.g., via high frequency and/or sensor signals).

[0037] In embodiments, an electrical system **400** may include and/or be connected to a power source **404**, such as a battery. An electrical connector assembly **100** may be configured to monitor the power source **404** and provide information about the power source **404** (e.g., temperature, peak current, voltage, impedance, etc.) to the ECU **402** and/or a remote location, which may provide efficient battery monitoring. The electrical connector assembly **100** may, for example and without limitation, provide the information via one or more conductors **204**, **208** of a cable **202**, **206** that may also be utilized for providing power from the power source **404** to other components, such as via PLC.

[0038] With embodiments, such as generally illustrated in FIG. 9, a method **900** of operating an electrical system **400** may include sensing an electrical characteristic (e.g., current, voltage, resistance, etc.) (block **902**) and/or a temperature associated with one or more conductors (e.g., the first conductor **204**) (block **904**). The method **900** may include comparing the sensed electrical characteristic and/or the sensed temperature to a specified or predetermined range (block **910**). The method **900** may include detecting and/or indicating an error if the sensed electrical characteristic and/or the sensed temperature is not within the specified or predetermined range (block **912**). For example, if the ECU **402** determines that the electrical characteristic (e.g., a current) is outside a

specified or predetermined range (e.g., below a minimum value of the predetermined range), the ECU **402** may determine that an error, such as excess leakage current, has occurred.

[0039] In embodiments, determining whether an electrical characteristic is within a specified or predetermined range (block **910**) may include compensating for temperature variations, as the resistance of a conductor **204, 208** may vary with temperature. For example and without limitation, the ECU **402** may utilize both a sensed electrical characteristic and a sensed temperature to determine whether an error has occurred (blocks **910, 912**).

[0040] With embodiments, an ECU **402** may be configured to detect an error (e.g., block **912**) if a temperature sensed via the temperature sensor **306** (block **904**), such as inside an electrical connector assembly **100**, is not within a specified or predetermined temperature range (block **910**). For example, if the first connector member **102** and the second connector member **104** are not properly connected, the improper connection may increase the temperature inside the electrical connector assembly **100**, and if the improper connection is not detected, a corresponding poor electrical connection may lead to a connection failure. The ECU **402** may be configured to detect the connection error (e.g., via a temperature increase of a current transmitted through a smaller total cross-sectional area), which may allow for the connection to be repaired or replaced prior to failure.

[0041] In embodiments, an ECU **402** may be configured to detect an error and which conductor **204, 208** of a cable **202, 206** is associated with the error. In some circumstances, the electrical characteristic sensor **304** may be configured to sense (e.g., directly) electrical characteristics of each conductor **204, 208**. Additionally or alternatively, the electrical characteristic sensor **304** may be configured to sense electrical characteristics of a cable **202, 206** as a whole, which may be less complex and/or less expensive than directly sensing electrical characteristics of each conductor **204, 208**. The ECU **402** may be configured to identify the conductor(s) **204, 208** associated with an error even if the electrical characteristic sensor **304** is not configured for directly sensing each conductor **204, 208**. For example and without limitation, an electrical connector assembly **100** may include one or more resistors **310** (see, e.g., FIG. 3). A respective resistor **310** may be electrically connected (e.g., in series) with some or all conductors **204, 208** of a cable **202, 206**. The resistors **310** may, for example, each include a unique resistance and/or signature. The resistances and/or signatures of the resistors **310** may be provided to and/or stored in the ECU **402** such that in the event of an error with a particular conductor **204, 208** (e.g., a disconnection/open), the ECU **402** may utilize the stored resistance and/or signature to identify the conductor **204, 208** associated with the error. For example and without limitation, if a conductor **204, 208** is severed, the ECU **402** may detect a change in resistance or signature that may correspond to the unique resistance or signature of the conductor **204, 208**, and the ECU **402** may identify that conductor **204, 208** as the severed conductor, which may facilitate efficient repair and/or replacement. In embodiments, determining whether sensed information is within a threshold (block **910**) may include determining which conductor **204, 208** of a cable **202, 206** the sensed information corresponds to (e.g., which conductor **204, 208** has an error). With embodiments, sensing an electrical characteristic (block **902**) may include applying optical time domain reflectometry (OTDR) for active health monitoring. OTDR may, for example, utilize the unique resistance and/or signature to identify conductors **204, 208**.

[0042] With embodiments, the one or more resistors **310** may include one or more of a variety of shapes, sizes, configurations, and/or materials. For example and without limitation, a resistor **310** may be configured as a through-hole resistor, a surface mounted resistor (e.g., mounted to the circuit board **118**), and/or a conductive material (e.g., copper) printed on a portion of a connector member **102, 104** (e.g., the first housing member **106** and/or the second housing member **112**).

[0043] With embodiments, an ECU **402** may be configured to predict errors in an electrical system **400** (block **908**), such as with an electrical connector assembly **100** and/or a cable **202, 206** connected thereto. For example and without limitation, the ECU **402** may be configured to monitor temperature, current, and/or leakage current over time, store the monitored/sensed information

(block **906**) and predict when a failure in the electrical system **400** is likely to occur (e.g., an expected failure time according to a rate of change) (block **908**).

[0044] Embodiments of an electrical system **400** may allow for certain components to be omitted, such as external circuit breakers and/or circuit interrupters that might otherwise be separately connected to a cable **202**, **206** or conductor **204**, **208**, as an electrical connector assembly **100** and/or an ECU **402** connected thereto may provide similar functionality. For example and without limitation, an electrical characteristic sensor **304** integrated with (e.g., without a separate housing) an electrical connector assembly **100** may provide current monitoring, which may allow for one or more external/central circuit breakers and/or interrupters to be omitted from the electrical system **400**. Additionally or alternatively, the electrical system **400** may be configured to respond to errors more quickly than conventional circuit breakers and/or interrupters that may require a relatively significant buildup of heat before switching/disconnecting.

[0045] While some embodiments of electrical connector assemblies **100** and connector members **102**, **104** are generally illustrated and described with one circuit board **118**, other embodiments of electrical connector assemblies **100** and connector members **102**, **104** may include more than one circuit board **118**.

[0046] In examples, an ECU (e.g., ECU **402**) may include an electronic controller and/or include an electronic processor, such as a programmable microprocessor and/or microcontroller. In embodiments, an ECU may include, for example, an application specific integrated circuit (ASIC). An ECU may include a central processing unit (CPU), a memory (e.g., a non-transitory computer-readable storage medium), and/or an input/output (I/O) interface. An ECU may be configured to perform various functions, including those described in greater detail herein, with appropriate programming instructions and/or code embodied in software, hardware, and/or other medium. In embodiments, an ECU may include a plurality of controllers. In embodiments, an ECU may be connected to a display, such as a touchscreen display.

[0047] Various examples/embodiments are described herein for various apparatuses, systems, and/or methods. Numerous specific details are set forth to provide a thorough understanding of the overall structure, function, manufacture, and use of the examples/embodiments as described in the specification and illustrated in the accompanying drawings. It will be understood by those skilled in the art, however, that the examples/embodiments may be practiced without such specific details. In other instances, well-known operations, components, and elements have not been described in detail so as not to obscure the examples/embodiments described in the specification. Those of ordinary skill in the art will understand that the examples/embodiments described and illustrated herein are non-limiting examples, and thus it can be appreciated that the specific structural and functional details disclosed herein may be representative and do not necessarily limit the scope of the embodiments.

[0048] Reference throughout the specification to “examples,” “in examples,” “with examples,” “various embodiments,” “with embodiments,” “in embodiments,” or “an embodiment,” or the like, means that a particular feature, structure, or characteristic described in connection with the example/embodiment is included in at least one embodiment. Thus, appearances of the phrases “examples,” “in examples,” “with examples,” “in various embodiments,” “with embodiments,” “in embodiments,” or “an embodiment,” or the like, in places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more examples/embodiments. Thus, the particular features, structures, or characteristics illustrated or described in connection with one embodiment/example may be combined, in whole or in part, with the features, structures, functions, and/or characteristics of one or more other embodiments/examples without limitation given that such combination is not illogical or non-functional. Moreover, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the scope thereof.

[0049] It should be understood that references to a single element are not necessarily so limited and may include one or more of such element. Any directional references (e.g., plus, minus, upper, lower, upward, downward, left, right, leftward, rightward, top, bottom, above, below, vertical, horizontal, clockwise, and counterclockwise) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of examples/embodiments.

[0050] Joinder references (e.g., attached, coupled, connected, and the like) are to be construed broadly and may include intermediate members between a connection of elements and relative movement between elements. As such, joinder references do not necessarily imply that two elements are directly connected/coupled and in fixed relation to each other. The use of “e.g.” in the specification is to be construed broadly and is used to provide non-limiting examples of embodiments of the disclosure, and the disclosure is not limited to such examples. Uses of “and” and “or” are to be construed broadly (e.g., to be treated as “and/or”). For example and without limitation, uses of “and” do not necessarily require all elements or features listed, and uses of “or” are inclusive unless such a construction would be illogical.

[0051] While processes, systems, and methods may be described herein in connection with one or more steps in a particular sequence, it should be understood that such methods may be practiced with the steps in a different order, with certain steps performed simultaneously, with additional steps, and/or with certain described steps omitted.

[0052] All matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative only and not limiting. Changes in detail or structure may be made without departing from the present disclosure.

[0053] It should be understood that an electronic control unit (ECU), a system, and/or a processor as described herein may include a conventional processing apparatus known in the art, which may be capable of executing preprogrammed instructions stored in an associated memory, all performing in accordance with the functionality described herein. To the extent that the methods described herein are embodied in software, the resulting software can be stored in an associated memory and can also constitute means for performing such methods. Such a system or processor may further be of the type having ROM, RAM, ROM and RAM, and/or a combination of non-volatile and volatile memory so that any software may be stored and yet allow storage and processing of dynamically produced data and/or signals.

[0054] It should be further understood that an article of manufacture in accordance with this disclosure may include a non-transitory computer-readable storage medium having a computer program encoded thereon for implementing logic and other functionality described herein. The computer program may include code to perform one or more of the methods disclosed herein. Such embodiments may be configured to execute via one or more processors, such as multiple processors that are integrated into a single system or are distributed over and connected together through a communications network, and the communications network may be wired and/or wireless. Code for implementing one or more of the features described in connection with one or more embodiments may, when executed by a processor, cause a plurality of transistors to change from a first state to a second state. A specific pattern of change (e.g., which transistors change state and which transistors do not), may be dictated, at least partially, by the logic and/or code.

Claims

1-20. (canceled)

21. An electrical connector assembly comprising: a first connector including a first housing with a first end and a second end, the first housing extending axially between the first end and the second end of the first housing, the first connector also including a first insulator, a circuit board, and first terminals, the first terminals being electrically connected with first conductors of a first cable, the

first conductors being routed into the first housing through the first end of the first housing and the first end of the first housing being sealed, the first housing surrounding the first insulator, the circuit board and the first terminals, the first insulator including a first portion and a second portion, the first and second portions of the first insulator supporting the first terminals with respect to each other, the first terminals extending axially through the circuit board and the first and second portions of the first insulator, wherein the circuit board is positioned axially between the first and second portions of the first insulator and defines a plurality of apertures through which the first terminals axially extend, wherein an electrical characteristic sensor is mounted on the circuit board, wherein the sensor is electrically connected to at least one of the first terminals by an electrical pathway that includes a trace on the circuit board, and wherein the electrical pathway directly electrically contacts the at least one of the first terminals; a second connector including a second housing with a first end and a second end, the second housing extending axially between the first end and the second end of the first housing, the second connector also including a second insulator and second terminals, the second terminals being electrically connected with second conductors of a second cable, the second conductors being routed into the second housing through the first end of the second housing and the first end of the second housing being sealed, the second terminals extending axially through the second insulator, the second housing surrounding the second insulator and the second terminals; and wherein the second ends of the first and second housings are configured to be coupled together causing the first terminals and the second terminals to be electrically connected to each other, and wherein a seal provides sealing between the first and second connectors when the first and second housings are coupled together.

22. The electrical connector assembly of claim 21, wherein the electrical characteristic sensor includes a voltage sensor and/or a current sensor.

23. The electrical connector assembly of claim 21, wherein the circuit board includes a current loop.

24. The electrical connector assembly of claim 23, wherein the current loop is electrically connected to the first terminals to provide an electrical connection between each conductor connected to respective first terminals and the electrical characteristic sensor such that at least one conductor provides power to the electrical characteristic sensor.

25. The electrical connector assembly of claim 21, wherein the circuit board includes a temperature sensor.

26. The electrical connector assembly of claim 25, wherein the circuit board, the electrical characteristic sensor, and/or the temperature sensor may be configured to draw power from and/or provide information/data to the first conductors via the first terminals.

27. The electrical connector assembly of claim 21, wherein the circuit board is configured for electrical connection at least one of the first conductors to provide power to the electrical characteristic sensor.

28. The electrical connector assembly of claim 21, wherein the circuit board is connected to the first insulator and the second insulator via an adhesive.

29. The electrical connector assembly of claim 21, wherein the plurality of apertures are plated through holes electrically connected with the traces.

30. The electrical connector assembly of claim 21, wherein the first terminals are disposed in the first insulator and the circuit board and extend through the plated through holes of the circuit board.

31. The electrical connector assembly of claim 21, wherein the circuit board includes pins configured to connect with the first and/or second terminals.

32. The electrical connector assembly of claim 21, wherein the second end of the first housing fits over the second end of the second housing.

33. The electrical connector assembly of claim 21, wherein an electronic control unit is connected with the circuit board, and wherein the electronic control unit is electrically connected with the electrical characteristic sensor and/or a temperature sensor of the circuit board via one or more first

conductors, one or more first terminals, and/or one or more traces of the circuit board.

34. The electrical connector assembly of claim 21, wherein the first housing and the second housing are cylindrical housings.

35. The electrical connector assembly of claim 21, the first housing including a first grommet sealing the first end of the first housing; where the second housing including a second grommet sealing the first end of the first housing, and wherein each of the first and second grommets include a plurality of apertures for receiving respective first and second conductors of the first and second connectors.

36. The electrical connector assembly of claim 35, wherein the second connector includes a circuit board.

37. The electrical connector assembly of claim 36, wherein the first and second grommets, the first and second insulators, and the circuit boards are cylindrical.

38. The electrical connector assembly of claim 36, wherein the second insulator includes a first portion and a second portion.

39. The electrical connector assembly of claim 38, wherein the circuit board of the second connector is positioned axially between the first and second portions of the second insulator and defines a plurality of apertures extending axially through the circuit board of the second connector.

40. An electrical connector assembly comprising: a first connector including a first housing with a first end and a second end, the first housing extending axially between the first end and the second end of the first housing, the first connector also including a first insulator, a circuit board and first terminals, the first terminals being electrically connected with first conductors of a first cable, the first conductors being routed into the first housing through the first end of the first housing and the first end of the first housing being sealed, the first housing surrounding the first insulator, the circuit board and the first terminals, the first insulator including a first portion and a second portion, the first and second portions of the first insulator supporting the first terminals in spaced-apart, parallel relation with respect to each other, the first terminals extending axially through the circuit board and the first and second portions of the first insulator, wherein the circuit board is positioned axially between the first and second portions of the first insulator and defines a plurality of apertures extending axially through the circuit board, wherein the plurality of apertures includes plated through holes that receive and electrically connect with the first terminals, wherein an electrical characteristic sensor is mounted on the circuit board, wherein the electrical characteristic sensor is electrically connected to the plated through holes by traces on the circuit board, and wherein the first terminals electrically connect with the electrical characteristic sensor by the traces and the plated through holes such that the first conductors electrically connect with the electrical characteristic sensor; a second connector including a second housing with a first end and a second end, the second housing extending axially between the first end and the second end of the first housing, the second connector also including a second insulator and second terminals, the second terminals being electrically connected with second conductors of a second cable, the second conductors being routed into the second housing through the first end of the second housing and the first end of the second housing being sealed, the second terminals extending axially through the second insulator and being supported in parallel spaced-part relation relative to each other by the second housing, the second housing surrounding the second insulator and the second terminals; and wherein the second ends of the first and second housings are configured to be coupled together causing the first terminals and the second terminals to be electrically connected to each other, and wherein a seal provides sealing between the first and second connectors when the first and second housing are coupled together.
